

Task: 9

15 Marks

Lab Task 1

Write a class **LocalPhone** that contains an attribute **phone** to store a local telephone number. The class contains member functions to input and display phone number. Write a child class **NatPhone** for national phone numbers that inherits **LocalPhone** class. It additionally contains an attribute to store **city code**. It also contains member functions to input and show the city code. Write another class **IntPhone** for international phone numbers that inherits **NatPhone** class. It additionally contains an attribute to store **country code**. It also contains member functions to input and show the country code. Write a main() to test the **IntPhone** class by creating instance of it.

Lab Task 2

Write a class Teacher that contains the attributes teacher **name**, **age**, and **address**. It also contains member functions to **input** and **display** its attributes. Write another class **Writer** that contains the attributes writer **name**, **address** and number of **books** written by him. It also has member functions to input and display its attributes. Write a third class **Scholar** that inherits both Teacher and Writer class. Create an instance of Scholar class and display the data.

Lab Task 3

Design a class named **PersonData** with the following member variables:

- lastName
- firstName
- address
- city
- state
- zip
- phone

Write the appropriate accessor and mutator functions for these member variables.

Next, design a class named **CustomerData**, which is derived from the **PersonData** class. The **customerData** class should have the following member variables:

- customerNumber
- mailingList

The **customerNumber** variable will be used to hold a unique integer for each customer. The **mailingList** variable should be a bool. It will be set to true if the customer wishes to be on a mailing list, or false if the customer does not wish to be on a mailing list. Write appropriate accessor and mutator functions for these member variables.

Demonstrate an object of the **CustomerData** class in a simple program.

PreferredCustomer Class

A retail store has a preferred customer plan where customers may earn discounts on all their purchases. The amount of a customer's discount is determined by the amount of the customer's cumulative purchases in the store.

- When a preferred customer spends \$500, he or she gets a 5% discount on all future purchases.
- When a preferred customer spends \$1,000, he or she gets a 6% discount on all future purchases.
- When a preferred customer spends \$1,500, he or she gets a 7% discount on all future purchases.
- When a preferred customer spends \$2,000 or more, he or she gets a 10% discount on all future purchases.

Design a class named **PreferredCustomer**, which is derived from the **CustomerData** class you created. The **PreferredCustomer** class should have the following member variables:

- purchasesAmount (a double)
- discountLevel (a double)

The **purchasesAmount** variable holds the total of a customer's purchases to date. The **discountLevel** variable should be set to the correct discount percentage, according to the store's preferred customer plan. Write appropriate member functions for this class and demonstrate it in a simple program.

Input Validation: Do not accept negative values for any sales figures.
